

NextGen Data Architects — UCU Analytics Platform

Technical overview: databases, data warehouse, tech stack, and architecture

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1. PostgreSQL databases in the system

The application uses **one PostgreSQL server** hosting multiple logical databases. In a full local/docker deployment there are **five** named databases:

#	Database name	Role
1	<code>ucu_sourcedb1</code>	Operational / academic-style source (students, courses, grades, enrollments, etc.). Simulates a
2	<code>ucu_sourcedb2</code>	Second source (e.g. HR: employees, positions, attendance). Separates administrative data from
3	<code>ucu_datawarehouse</code>	Gold analytical store: star schema (dimensions + facts), primary target for dashboards
4	<code>ucu_rbac</code>	Security and application state: app users, profiles, workspace state, audit logs. Keeps auth data
5	<code>airflow_meta</code>	Apache Airflow metadata (DAG runs, connections). Created by <code>postgres-init</code> for orchestr

Note: Docker init may also create schemas named like the source DBs inside a single database for HR tables; the Python app is configured for separate database names (see `backend/config.py` / `backend/config/connection.py`). Align deployment scripts with how you create databases.

2. Total number of data records

There is **no single static row count** checked into the repo: volumes depend on `setup_databases.py` runs (randomized ranges), synthetic CSV/XLSX packages, and ETL executions. Rough design targets include **1000+ rows per major source table** in generated setups, with facts in the warehouse matching ETL loads.

How to obtain exact totals in your environment:

- Warehouse facts: run `backend/scripts/print_fact_counts.py` (fact tables).
- Any table: `SELECT COUNT(*) FROM <table>;` in each database.
- Medallion files: sum row counts of the **latest** bronze/silver Parquet snapshot only (multiple timestamps exist; do not sum all files or you double-count).

3. Nature of the data warehouse and rationale

Medallion architecture (Bronze → Silver → Gold)

- **Bronze:** raw or lightly typed extracts (Parquet under `backend/data/bronze`).
- **Silver:** cleansed, conformed datasets (Parquet under `backend/data/silver`).
- **Gold:** **PostgreSQL** in `ucu_datawarehouse` — star schema (dimensions + fact tables) for reporting, KPIs, and ML features.

Why these databases?

- **Two sources (DB1 + DB2):** models real separation between academic and HR/admin systems without merging operational concerns prematurely.
- **Warehouse (PostgreSQL):** strong SQL, constraints, and tooling for star schemas; single place for analytics and role-scoped APIs.
- **RBAC database:** isolates credentials, profiles, and audit trails from analytical facts.
- **Airflow DB:** standard pattern for workflow metadata.

4. Tech stack

Layer	Technologies
Frontend	React 18, React Router, Chakra UI, Tailwind-related utilities, Axios, ECharts, Recharts, Plotly, Monaco Editor, j
Backend API	Flask 3, Flask-JWT-Extended, Flask-CORS, Gunicorn, Werkzeug, SQLAlchemy, psycpg2
Data & ML	pandas, NumPy, scikit-learn, pyarrow (ETL Parquet)
Reports	ReportLab, openpyxl
Orchestration	Apache Airflow (DAGs under <code>airflow/dags</code>)
Data stores	PostgreSQL 16 (Docker image in compose)
DevOps	Docker Compose, Render blueprint (<code>render.yaml</code>)

5. Database tables (inventory)

Counts and table lists below are pulled from the running PostgreSQL instance (non-system schemas only).

ucu_sourcedb1 — 10 tables

Schema.Table	Importance
public.attendance	Operational/source table used as ETL input.
public.courses	Operational/source table used as ETL input.
public.departments	Operational/source table used as ETL input.
public.enrollments	Operational/source table used as ETL input.
public.faculties	Operational/source table used as ETL input.
public.grades	Operational/source table used as ETL input.
public.lecturers	Operational/source table used as ETL input.
public.programs	Operational/source table used as ETL input.
public.student_fees	Operational/source table used as ETL input.
public.students	Operational/source table used as ETL input.

ucu_sourcedb2 — 9 tables

Schema.Table	Importance
public.assets	Operational/source table used as ETL input.
public.contracts	Operational/source table used as ETL input.
public.employee_attendance	Operational/source table used as ETL input.
public.employees	Operational/source table used as ETL input.
public.maintenance_records	Operational/source table used as ETL input.
public.payroll	Operational/source table used as ETL input.
public.positions	Operational/source table used as ETL input.
public.purchase_orders	Operational/source table used as ETL input.
public.suppliers	Operational/source table used as ETL input.

ucu_datawarehouse — 26 tables

Schema.Table	Importance
public.dim_app_user	Dimension table used to slice facts (filters/labels).
public.dim_course	Dimension table used to slice facts (filters/labels).
public.dim_date	Dimension table used to slice facts (filters/labels).
public.dim_department	Dimension table used to slice facts (filters/labels).
public.dim_employee	Dimension table used to slice facts (filters/labels).
public.dim_faculty	Dimension table used to slice facts (filters/labels).
public.dim_high_school	Dimension table used to slice facts (filters/labels).
public.dim_program	Dimension table used to slice facts (filters/labels).

public.dim_semester	Dimension table used to slice facts (filters/labels).
public.dim_student	Dimension table used to slice facts (filters/labels).
public.dim_time	Dimension table used to slice facts (filters/labels).
public.fact_academic_performance	Fact table powering KPIs, charts, and analytics.
public.fact_attendance	Fact table powering KPIs, charts, and analytics.
public.fact_enrollment	Fact table powering KPIs, charts, and analytics.
public.fact_grade	Fact table powering KPIs, charts, and analytics.
public.fact_grades_summary	Fact table powering KPIs, charts, and analytics.
public.fact_payment	Fact table powering KPIs, charts, and analytics.
public.fact_progression	Fact table powering KPIs, charts, and analytics.
public.fact_sponsorship	Fact table powering KPIs, charts, and analytics.
public.fact_student_high_school	Fact table powering KPIs, charts, and analytics.
public.fact_transcript	Fact table powering KPIs, charts, and analytics.
ucu_sourcedb1.departments	Source-mirror schema stored in warehouse DB (HR/admin operational tables).
ucu_sourcedb1.faculties	Source-mirror schema stored in warehouse DB (HR/admin operational tables).
ucu_sourcedb2.employee_attendance	Source-mirror schema stored in warehouse DB (HR/admin operational tables).
ucu_sourcedb2.employees	Source-mirror schema stored in warehouse DB (HR/admin operational tables).
ucu_sourcedb2.positions	Source-mirror schema stored in warehouse DB (HR/admin operational tables).

ucu_rbac — 14 tables

Schema.Table	Importance
public.app_users	Application users + roles (login source of truth).
public.assigned_query_visualizations	Analyst SQL workspace assets (visualizations/feedback).
public.audit_logs	Audit trail for admin/security actions.
public.dashboard_role_access	Dashboards + access control for roles/users.
public.dashboard_user_access	Dashboards + access control for roles/users.
public.dashboards	Dashboards + access control for roles/users.
public.leave_requests	RBAC/admin support table.
public.page_config	Dashboards + access control for roles/users.
public.role_current_dashboard	Dashboards + access control for roles/users.
public.staff_course_assignments	RBAC/admin support table.
public.user_profiles	User profile metadata for UI.
public.user_state	Persisted per-user UI/workspace state.
public.viz_feedback	Analyst SQL workspace assets (visualizations/feedback).
public.viz_feedback_replies	Analyst SQL workspace assets (visualizations/feedback).

airflow_meta — 48 tables

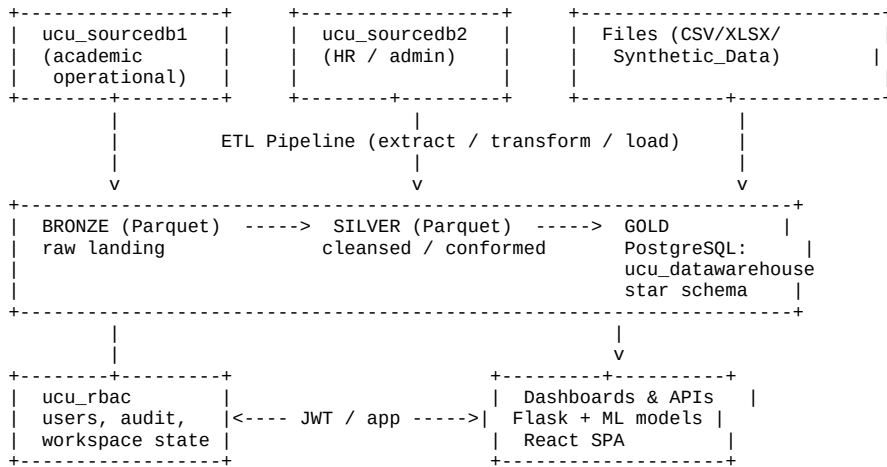
Schema.Table	Importance
public.ab_permission	Airflow metadata table (runs/tasks/schedules).

public.ab_permission_view	Airflow metadata table (runs/tasks/schedules).
public.ab_permission_view_role	Airflow metadata table (runs/tasks/schedules).
public.ab_register_user	Airflow metadata table (runs/tasks/schedules).
public.ab_role	Airflow metadata table (runs/tasks/schedules).
public.ab_user	Airflow metadata table (runs/tasks/schedules).
public.ab_user_role	Airflow metadata table (runs/tasks/schedules).
public.ab_view_menu	Airflow metadata table (runs/tasks/schedules).
public.alembic_version	Airflow metadata table (runs/tasks/schedules).
public.callback_request	Airflow metadata table (runs/tasks/schedules).
public.connection	Airflow metadata table (runs/tasks/schedules).
public.dag	Airflow metadata table (runs/tasks/schedules).
public.dag_code	Airflow metadata table (runs/tasks/schedules).
public.dag_owner_attributes	Airflow metadata table (runs/tasks/schedules).
public.dag_pickle	Airflow metadata table (runs/tasks/schedules).
public.dag_priority_parsing_request	Airflow metadata table (runs/tasks/schedules).
public.dag_run	Airflow metadata table (runs/tasks/schedules).
public.dag_run_note	Airflow metadata table (runs/tasks/schedules).
public.dag_schedule_dataset_alias_reference	Airflow metadata table (runs/tasks/schedules).
public.dag_schedule_dataset_reference	Airflow metadata table (runs/tasks/schedules).
public.dag_tag	Airflow metadata table (runs/tasks/schedules).
public.dag_warning	Airflow metadata table (runs/tasks/schedules).
public.dagrun_dataset_event	Airflow metadata table (runs/tasks/schedules).
public.dataset	Airflow metadata table (runs/tasks/schedules).
public.dataset_alias	Airflow metadata table (runs/tasks/schedules).
public.dataset_alias_dataset	Airflow metadata table (runs/tasks/schedules).
public.dataset_alias_dataset_event	Airflow metadata table (runs/tasks/schedules).
public.dataset_dag_run_queue	Airflow metadata table (runs/tasks/schedules).
public.dataset_event	Airflow metadata table (runs/tasks/schedules).
public.import_error	Airflow metadata table (runs/tasks/schedules).
public.job	Airflow metadata table (runs/tasks/schedules).
public.log	Airflow metadata table (runs/tasks/schedules).
public.log_template	Airflow metadata table (runs/tasks/schedules).
public.rendered_task_instance_fields	Airflow metadata table (runs/tasks/schedules).
public.serialized_dag	Airflow metadata table (runs/tasks/schedules).
public.session	Airflow metadata table (runs/tasks/schedules).
public.sla_miss	Airflow metadata table (runs/tasks/schedules).
public.slot_pool	Airflow metadata table (runs/tasks/schedules).
public.task_fail	Airflow metadata table (runs/tasks/schedules).
public.task_instance	Airflow metadata table (runs/tasks/schedules).
public.task_instance_history	Airflow metadata table (runs/tasks/schedules).
public.task_instance_note	Airflow metadata table (runs/tasks/schedules).

public.task_map	Airflow metadata table (runs/tasks/schedules).
public.task_outlet_dataset_reference	Airflow metadata table (runs/tasks/schedules).
public.task_reschedule	Airflow metadata table (runs/tasks/schedules).
public.trigger	Airflow metadata table (runs/tasks/schedules).
public.variable	Airflow metadata table (runs/tasks/schedules).
public.xcom	Airflow metadata table (runs/tasks/schedules).

6. Database architecture (logical)

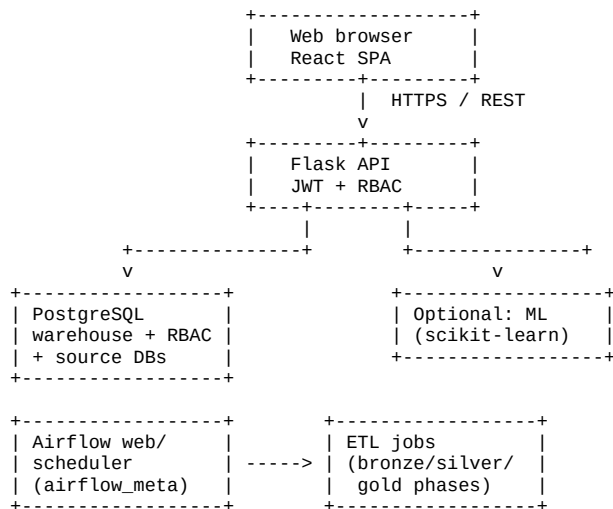
High-level data flow from sources through medallion layers to consumption.



Orchestration: Apache Airflow --> metadata DB: airflow_meta

7. System architecture (application)

End-to-end view of clients, API, and services.



This document was generated automatically. Tune database creation and record counts to match your deployment.